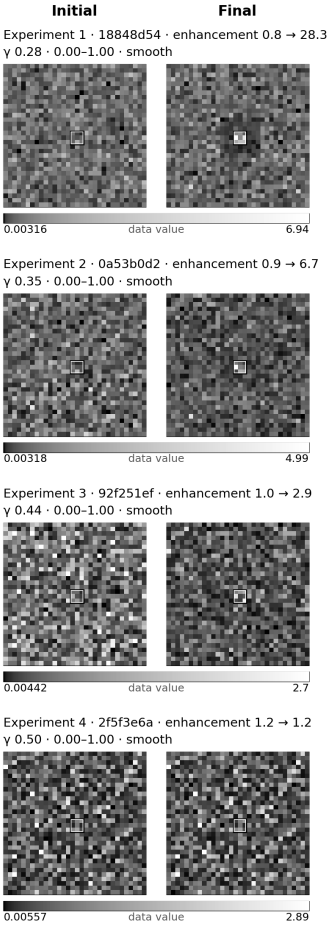
Macropixel scale and focusing performance

Research question | How does control-block size affect validated focusing enhancement?

On the fixed simulated bench, validated enhancement fell from 28.3 at macropixel size 1 to 1.16 at size 8 in this four-run comparison. Enhancement is focus mean divided by the equally weighted mean of four background-region means; all runs held seed, regions and absolute iteration budget fixed.



Initial / Final heatmaps; macropixel sizes 1, 2, 4, 8. Grayscale tuning changes appearance, not measurements.

COMPARISON | common settings: fixed regions; search seed and iteration budget held constant

Macropixel size	Final enhancement	Iterations	Camera shots
1	28.268	1000 iterations	3 / search; 5 validation
2	6.716	1000 iterations	3 / search; 5 validation
4	2.873	1000 iterations	3 / search; 5 validation
8	1.160	1000 iterations	3 / search; 5 validation

VERDICT AND NEXT EXPERIMENT

This is fair at equal absolute budget and fixed measurement regions, but not per control block: sizes 1 and 2 average 3.9 and 15.6 iterations per block, respectively. Wall time and total camera-shot counts were not recorded; the fine-grid outcome is budget-limited, and one seed cannot establish robustness. Validated enhancement decreased across the selected runs, from 28.268 (size 1) to 1.160 (size 8); each used 1000 iterations, 3 camera shots per search measurement and 5 fresh validation shots. Next: repeat macropixel size 1 with search seed 2, holding other settings fixed, to test search-trajectory sensitivity.

Simulation; relative intensity units. Grayscale tuning is presentational and does not change the measurements.